Task 1 — Ethernet cable vs Fiber optic cable

The main difference is how they carry data. Ethernet cable normally carries electrical signals through copper wires, while fiber optic cable carries data as pulses of light. Both are useful, but the best choice depends on distance, speed, interference and cost.

Point	Ethernet (Copper)	Fiber Optic
Signal	Electrical signals through copper pairs	Light signals through glass/plastic fiber
Distance & interference	Typical Ethernet copper link is up to about 100 m; can be affected by electromagnetic interference	Good for long links and resistant to electromagnetic interference
Cost & use	Usually cheaper and easy to terminate with an RJ45 plug; common in homes and offices	Optical modules and specialist termination tools may add cost; common in data centers and campus backbones

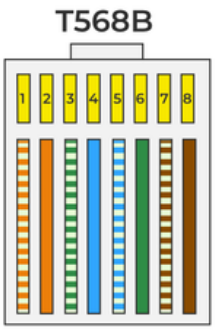
Real-world examples: Ethernet is suitable for connecting a home Wi-Fi router to a PC or connecting the 10 PCs inside a small gaming café. Fiber is suitable for connecting different buildings on a college campus or linking network racks over longer distances.

Task 2 — T568B straight-through Ethernet cable

A straight-through cable uses the same wiring standard at both ends. With T568B, hold each RJ45 plug with the gold contacts facing you and the clip away from you; use the pin order below from left to right. Keep the twisted pairs as close to the plug as practical and check the order before crimping.

Pin	Wire colour	Pair / purpose
1	Orange / white	Pair 2
2	Orange	Pair 2
3	Green / white	Pair 3
4	Blue	Pair 1
5	Blue / white	Pair 1
6	Green	Pair 3
7	Brown / white	Pair 4
8	Brown	Pair 4

Memory order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
Simple diagram: T568B at both ends — Pin 1 to Pin 1, Pin 2 to Pin 2, and so on through Pin 8. This is why it is called straight-through.



Task 3 — Steps to attach an RJ45 connector to a CAT6 cable

1. Measure and cut: Measure the cable to the required length and cut it neatly with a cable cutter.

2. Strip the jacket: Use the stripper on the crimper/tool to remove a short section of outer jacket without nicking the inner conductors.

3. Untwist and arrange: Separate the four twisted pairs and arrange the wires in T568B order: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.

4. Trim evenly: Straighten the conductors gently and trim the ends evenly so they can reach the front of the RJ45 plug.

5. Insert into plug: Slide all eight wires into the correct channels of a CAT6-compatible RJ45 plug. Check that the jacket also goes slightly inside the plug for strain relief.

6. Crimp: Place the plug into the RJ45 crimper and squeeze fully so the contacts press into the wires and the plug locks onto the cable.

7. Test: Use a cable tester to check that pins 1–8 connect in the correct order. If a wire is open, crossed or swapped, cut off the plug and terminate again.

Safety and quality tip: Do not strip too much jacket or untwist the pairs farther than necessary. Use a CAT6-rated plug suitable forthe cable type(solid or stranded) and check the testerresult before installing it.

Task 4 — Cable choice for a gaming café with 10 PCs

I would use CAT6 Ethernet copper cables to connect each of the 10 PCs to a network switch, because the cable runs inside a small café are usually short and Ethernet is affordable and easy to maintain. I would use an RJ45 crimper to make the cable ends and a cable tester to confirm that every wire is in the correct order. Fiber is more useful if the café needs a long link to another building or a network room far away; in that case, I would use fiber with compatible optical equipment and a fiber cleaver for termination. For a normal single-room gaming café, CAT6 gives a practical balance of speed, budget and simplicity.

Conclusion: Ethernet copper is a practical choice for short device connections, while fiber is better suited to long-distance and interference-resistant links. Correct T568B termination and cable testing help avoid network faults.